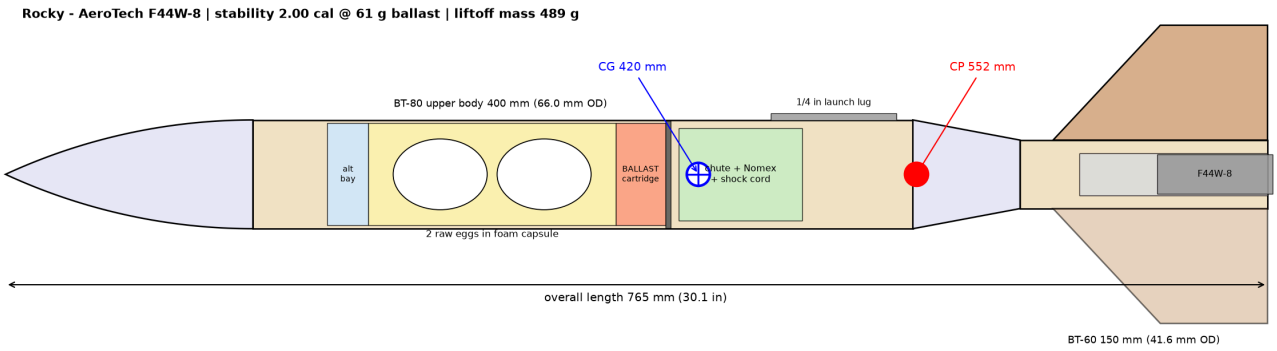
ARC 2027 — AeroTech Design "Rocky" (F44W-8, adjustable ballast)

Design and simulation report for Krish's team. Prepared October 2, 2026 for the **DARS Gunter, TX test flight on Saturday, Oct 17, 2026** (test flight, not scored). OpenRocket 24.12 (run headless). OpenRocket file: `ARC AeroTech Design - Rocky.ork` (md5 `c5cd42ccdd59bc9534dc760936b97bb5`).



1. Summary

- **This is a new rocket designed from scratch.** It has a 66 mm (BT-80) payload section, a molded plastic transition, and a short 41.6 mm (BT-60) booster with three large balsa fins. Overall length is **765 mm (30.1 in)**.
- **Motor: AeroTech F44W-8.** This is a 24 × 70 mm single-use motor (ARC list 41.5 N·s; OpenRocket curve 42.2 N·s, avg 51.4 N, 0.82 s burn). It is on the 2027 ARC certified list and not crossed out. Buy motors made **after Aug 2024** (the list notes that older F44W stock is not recommended).
- **Adjustable ballast cartridge:** a 3D-printed PETG cup filled with modeling clay. It sits on the payload bulkhead directly under the egg capsule. Its center is at 385 mm from the nose tip, which is only **about 35–40 mm (0.5–0.6 caliber) forward of the CG**. Because the ballast sits forward of the CG, adding ballast can only *raise* stability, and only by about 0.0012 cal per gram.
  - Static stability is **1.94 cal with the cartridge empty (0 g)** and **1.99–2.03 cal across every ballast setting on the card (41–74.5 g)**.
  - Even 140 g would only give 2.09 cal. **Stability never drops below 1.94 cal at any ballast amount.**
- **Hitting 800 ft:** using the ballast card, mean apogee is 800 ft at every wind (0/5/10/15/20 mph) and every temperature tested (40–90 °F).
  - On 30 *fresh* random seeds per condition, the 35 validated means fall between **796.7 and 803.6 ft**.
  - Liftoff mass is **469–503 g**, leaving at least 147 g of margin under the 650 g limit.
- **Flight time:** with the **18.5 in (470 mm) ripstop chute**, mean flight time is **38.19–39.54 s at 50–80 °F** in all winds (inside the 37–40 s window).
  - At 60–80 °F, the expected range for Oct 17, it is 38.19–39.26 s.
  - On cold, windy days (≤50 °F with ≥15 mph wind), use the 18 in chute instead (Section 7.3).
- **Rail exit speed is 66.6–71.3 ft/s** on a 6 ft rod (79.6–83.4 ft/s on an 8 ft rod), well above 45 ft/s.
  - Rail-exit stability is 2.06 cal in calm air. It falls in crosswind because the wind tilts the airflow at rail exit: 1.65 / 1.37 / 1.17 / 1.04 cal at 5 / 10 / 15 / 20 mph.
  - **These values are below 1.5 cal at 10 mph and above (flagged).** An 8 ft rod raises them to 1.72 / 1.48 / 1.27 / 1.14 cal.

- **Deployment** happens about 1.9 s after apogee (8 s delay vs. an optimum of about 6.1 s), at **48 ft/s in calm air and about 60–65 ft/s in 10–20 mph wind**. That is the main compromise of this motor (Section 4).
    - Descent rate is **about 23–26 ft/s**. Drift is about 9 / 150 / 300 / 460 / 650 ft at 0 / 5 / 10 / 15 / 20 mph.
- **Total OpenRocket flight simulations run: 24,970** (breakdown in Section 13).

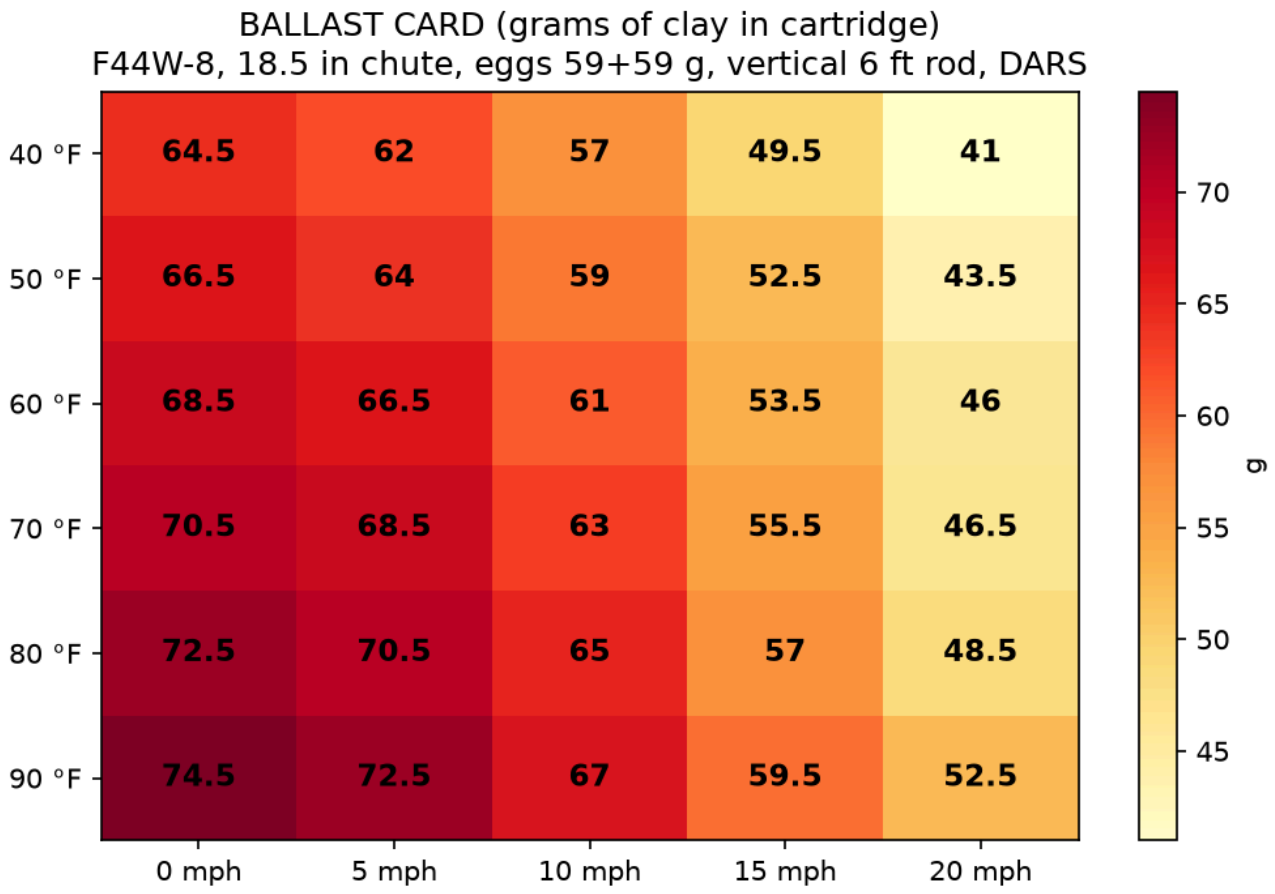
Ballast card (grams of clay in the cartridge), 18.5 in chute, vertical 6 ft rod, 59 g + 59 g eggs

Temp \ Wind	0 mph	5 mph	10 mph	15 mph	20 mph
40 °F (ext.)	**64.5**	**62**	**57**	**49.5**	**41**
50 °F	**66.5**	**64**	**59**	**52.5**	**43.5**
60 °F	**68.5**	**66.5**	**61**	**53.5**	**46**
70 °F	**70.5**	**68.5**	**63**	**55.5**	**46.5**
80 °F	**72.5**	**70.5**	**65**	**57**	**48.5**
90 °F (ext.)	**74.5**	**72.5**	**67**	**59.5**	**52.5**

Baseline saved in the .ork file: 61 g (60 °F, 10 mph).

Egg correction: weigh both eggs together. Then **add (118 g − egg total) to the card value.** For example, eggs totaling 112 g → add 6 g; eggs totaling 124 g → remove 6 g. This was checked in simulation: compensated egg swaps return the apogee to 800 ft (Section 10).

Other temperatures: between rows, interpolate linearly (about +0.2 g per °F).



2. Rules compliance checklist (2027 ARC Rules §4.3–4.10 + NAR)

Requirement	Rule	This design	Status
Gross liftoff mass	≤ 650 g	469–503 g across the ballast card (≤ 511 g even with 63 g + 63 g eggs and no correction)	✅
Overall length	≥ 650 mm (25.6 in)	765 mm	✅
Two body-tube diameters; one tube ≥ 66 mm (2.6 in) OD and ≥ 300 mm long	§4.3	BT-80: 66.04 mm (2.600 in) OD × **400 mm** long	✅ (**measure the actual tube with calipers; it must be ≥ 66.0 mm**)
Other tube differs by ≥ 9 mm (0.35 in) in OD	§4.3	BT-60: 41.58 mm OD (24.5 mm smaller)	✅
Single stage; F class or lower; ≤ 80 N·s total; certified motor on the ARC list	§4.3 / App. 7	One AeroTech F44W-8, 41.5 N·s (ARC list); not crossed out	✅ (buy post-Aug-2024 production)
Motor positively retained, removable after flight	§4.3	3D-printed PETG screw-on retainer (no metal) + paper engine block	✅
No pyrotechnics other than the motor	§4.3	Motor ejection charge only	✅
Parachute recovery; all parts tethered together	§4.3 / 4.8	18.5 in nylon chute. Payload and booster joined by Kevlar + elastic. Nose cone taped to the upper tube with a Kevlar thread backup tether	✅
Not a commercial ARC/egg kit	§4.3	Scratch design from generic parts	✅
2 raw eggs, 55–63 g, ≤ 45 mm diameter, fully enclosed, no cracks	§4.4	Two-cavity foam capsule inside the 65 mm ID BT-80	✅ (survival must be proven in flight)
One approved altimeter (Jolly Logic AltimeterOne/Two or PerfectFlite Pnut/Firefly), unmodified	§4.6	AltimeterTwo (about 10 g) in a foam puck; 3 × 1.5 mm static vent holes	✅
Team entry number on rocket; paint/coating on paper or wood for Finals	§4.3	Primer/paint allowance included in mass (assumed 8 g + 8 g)	⚠️ (write the number on the rocket)
≥ 6 ft rigid guidance; rod ≥ 1/4 in (local); rails only at Finals	§4.10	1/4 in launch lug for DARS. **For Finals, add 3D-printed 1010 rail guides and re-simulate**	✅ (local) / ⚠️ (Finals)
NAR: no launch in wind > 20 mph	Safety code	Ballast card covers 0–20 mph	✅
Scoring: 800 ft; 37–40 s	§4.5 / 4.6	Mean 800 ft; mean flight time 38.2–39.5 s at 50–80 °F	✅ (simulated)

3. Design overview and materials

- **Airframe:** kraft cardboard body tubes. BT-80 upper body (66.04 OD / 64.97 ID) × 400 mm; BT-60 lower body (41.58 OD / 40.51 ID) × 150 mm.
- **Nose and transition:** molded polystyrene plastic. Ogive nose 150 mm long + 45 mm shoulder; conical transition 65 mm long.
- **Motor mount:** 24 mm kraft phenolic motor mount tube (114 mm) and 3.2 mm kraft phenolic centering rings. The fin tabs glue to the motor tube.
- **Fins:** three 3/16 in (4.76 mm) balsa fins, through-the-wall, with rounded edges.
- **Recovery:** ripstop nylon chute, Nomex protector, flat elastic upper shock cord, Kevlar lower shock cord tied around the motor tube.
- **No metal anywhere in the airframe.** The cords are tied through holes in the bulkhead (no eyebolts), the motor retainer is printed PETG, and the ballast is clay.
- **Why this shape:** a long, light upper section that stays close to the CG, plus a short, large-finned booster.
    - Geometry tests (14 variants, Section 13) showed that a shorter booster with bigger fins gives the best rail-exit stability in wind at the same 2.0 cal static margin.
    - Example (F44W, 20 mph, 6 ft rod): booster 220 mm with 110 mm root fins → 0.87 cal at rail exit; booster 140–160 mm with 130–160 mm root fins → 1.02–1.13 cal.

4. Motor choice

4.1 Stage 1 screen: every available AeroTech / Quest-AeroTech E–F motor on the same new airframe

Method:

- Each motor got its own fins, resized so stability was 2.0 cal with 0 g ballast. Simulations used DARS conditions at 60 °F with the stage-1 airframe (220 mm booster, 18 in chute).
- Ballast needed for 800 ft was solved at 0 and 20 mph. Every listed delay was tried, plus RDDT-drilled delays for reloads, with ballast tuned for each.
- The row shown for each motor is the delay with the lowest deployment speed. "Mass" is liftoff mass at the 0 mph ballast.
- Crossed-out motors were excluded, and partial motors were limited to their available delays (F25W: 4 and 6; F26FJ: 6; F52C: 5 and 8; F67C: 6 and 9).

Motor (curve)	Size	Reload	I (N·s, OR)	Avg F (N)	Ballast 0 / 20 mph (g)	Best delay	Deploy 0 / 20 mph (ft/s)	Liftoff mass (g)	Rail exit (ft/s)	Rail stab 20 mph	Verdict
E16W (0)	29x124	Y	37.7	20.8	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
E20W (0)	24x65	N	34.7	22.1	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
E23T (0)	29x124	Y	35.3	23.7	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
E24C (0)	29x110	N	37.5	22.9	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
E26W (0)	24x70	N	27.6	23.5	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
E30T (0)	24x70	N	33.7	34.9	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
E30T (1)	24x70	N	39.5	33.9	43 / 19	7 (listed)	34 / 40	463	61.8	0.92	OR curve (39.5 N·s) does not match the certified 33.6 N·s; data unreliable, rejected
E30T (2)	24x70	N	39.5	33.9	44 / 19	7 (listed)	35 / 42	463	61.9	0.87	OR curve (39.5 N·s) does not match the certified 33.6 N·s; data unreliable, rejected
E35W (0)	24x113	N	39.3	35.3	0 g ballast gives < 800 ft	–	–	–	–	–	Too weak: cannot reach 800 ft at 20 mph even with 0 g ballast
F20W (0)	29x73	N	60.6	26.5	250 / 196	4 (listed)	37 / 59	702	49.1	0.96	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F20W (1)	29x73	N	51.7	22.3	147 / 105	4 (listed)	37 / 55	606	48.6	0.80	Slow (49 ft/s rail exit), 605 g, 4 s delay deploys early
F25W (0)	29x98	N	70.8	29.6	344 / 287	6 (listed)	23 / 57	820	50.1	1.12	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F26FJ (0)	29x98	N	62.5	25.2	239 / 196	6 (listed)	19 / 63	718	41.3	0.99	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F27R (0)	29x83	N	49.5	24.1	122 / 88	4 (listed)	37 / 49	580	45.9	0.78	Rail exit 46 ft/s (marginal), rail stab 0.78 cal, deploys 37–49 ft/s

Motor (curve)	Size	Reload	I (N·s, OR)	Avg F (N)	Ballast 0 / 20 mph (g)	Best delay	Deploy 0 / 20 mph (ft/s)	Liftoff mass (g)	Rail exit (ft/s)	Rail stab 20 mph	Verdict
F30FJ (0)	24x90	N	47.1	30.4	110 / 75	6 (listed)	6 / 32	557	50.5	0.85	Finalist (stage 2)
F32T (0)	24x90	N	56.7	34.2	227 / 177	6 (listed)	9 / 42	667	54.2	0.99	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F32T (1)	24x90	N	56.9	34.6	228 / 184	6 (listed)	7 / 39	670	55.2	1.01	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F35W (0)	24x95	Y	57.6	40.7	225 / 188	6 (drilled)	3 / 36	688	54.1	1.01	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F37W (0)	29x99	Y	50.7	32.6	111 / 79	6 (listed)	4 / 28	598	47.7	0.72	Finalist (stage 2)
F40W (0)	29x124	Y	78.1	40.3	397 / 398	6 (drilled)	2 / 52	904	49.3	1.31	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F41W (0)	24x114	N	46.3	39.5	112 / 79	5 (listed)	37 / 48	550	49.3	0.87	Finalist (stage 2)
F42T (0)	29x83	N	55.9	42.8	217 / 183	8 (listed)	52 / 73	671	60.6	1.04	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F44W (0)	24x70	N	42.2	51.4	78 / 59	8 (listed)	49 / 64	502	69.2	0.92	**SELECTED (F44W-8)**: best rail exit, lightest, smallest ballast range
F50T (0)	29x98	N	68.7	54.5	351 / 345	6 (listed)	6 / 41	820	57.1	1.16	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F50T (1)	29x98	N	76.3	55.6	449 / 371	6 (listed)	9 / 46	913	56.6	1.24	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F51NT (0)	24x70	Y	52.9	53.5	184 / 150	6 (drilled)	6 / 26	643	65.2	1.06	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F52C (0)	29x112	N	66.3	52.8	324 / 294	5 (listed)	44 / 56	784	52.1	1.12	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F52T (0)	29x124	Y	73.0	53.6	364 / 303	6 (listed)	9 / 41	866	60.6	1.20	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F62T (0)	29x89	Y	57.6	58.0	208 / 192	6 (listed)	11 / 30	696	58.5	1.09	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F62T (1)	29x89	Y	51.0	51.3	130 / 101	6 (listed)	7 / 25	618	59.5	0.92	Two OR curves disagree (51 vs 57.6 N·s), so uncertainty is too high; 618 g
F63R (0)	24x95	Y	49.5	62.5	147 / 128	6 (drilled)	10 / 24	605	68.7	1.06	Finalist (stage 2)
F67C (0)	29x112	N	77.4	55.2	447 / 392	6 (listed)	13 / 49	912	50.7	1.23	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)
F67W (0)	29x89	N	62.1	73.3	295 / 269	6 (listed)	14 / 30	754	65.9	1.22	Too much impulse: needs so much ballast that liftoff mass ≥ 620 g (little or no margin)

4.2 Stage 2: five finalists, full wind × temperature matrix

Method:

- Final airframe geometry; fins resized for 2.0 cal at the mid ballast; chute resized for about 38.5 s.
- Ballast tuned for 800 ft at 5 winds × 4 temperatures (50/60/70/80 °F), then each condition validated on 10 fresh seeds.

Finalist	Chute (in)	Ballast range (g)	Stability range (cal)	Stability at 0 g	Liftoff mass max (g)	Mean apogee range (ft)	Mean flight time range (s)	Max deploy (ft/s)	Min rail exit (ft/s)	Worst rail stab at 20 mph
F44W-8	18	45.5–74.5	1.989–2.022	1.914	501	796–806	37.37–38.85	65	66.8	0.96
F30FJ-6	18	59–107.5	1.975–2.039	1.865	557	786–807	37.30–40.18	38	47.0	0.79
F63R-6	19.5	111.5–146.5	1.975–2.018	1.778	609	798–807	37.79–39.79	24	66.0	0.89
F41W-5	18	60–105	1.987–2.040	1.887	545	791–806	37.14–39.37	50	48.3	0.81
F37W-6	18.5	61–106.5	1.955–2.036	1.811	598	787–812	36.97–39.36	35	46.2	0.70

4.3 Why F44W-8

- **Best "any-weather" behavior.**
  - It has the highest rail-exit speed of all feasible motors: 67–71 ft/s vs. 46–50 ft/s for the F30FJ, F37W and F41W. Faster rail exit means a smaller wind angle at rail exit and the best windy rail-exit stability.
  - It has the tightest flight-time spread of the finalists and validated means closest to 800 ft.
- **Lightest and lowest-risk build.**
  - Maximum liftoff mass is about 503 g, leaving about 147 g of margin.
  - It needs the smallest ballast amount (41–75 g) and the smallest ballast range, so stability stays at 1.99–2.03 cal.
  - It is a single-use motor: no reload case, no delay drilling, cheap, and widely stocked.
- **Main compromise: the 8 s delay is about 1.9 s longer than optimum** (the optimum is about 6.1 s; F44W only comes with 4 or 8 s).
  - The chute opens about 55–65 ft below apogee at 48 ft/s in calm air and 60–65 ft/s in wind. OpenRocket's instant-inflation estimate of the opening load is 2.7 g (calm) to 8.7 g (20 mph).
  - This **does not affect the altimeter score**, because apogee has already been recorded.
  - The 4 s delay would deploy about 1 s *before* apogee at about 68 ft/s. That is worse, so it was rejected.
- **Runner-up: F30FJ-6.** Its deployment timing is excellent (5–38 ft/s), but rail exit is only 47–50 ft/s, windy rail stability is 0.79 cal, mass is 557 g, and its flight time reached 40.18 s at 50 °F / 20 mph in stage 2.
- **F63R-6 (reload, drilled):** fast rail exit, but it needs 112–147 g of ballast (609 g liftoff), reload hardware and an RDDT-drilled delay. Its stability with an empty cartridge would be 1.78 cal.

5. Parts / build list (OpenRocket model, baseline ballast 61 g)

- Positions are measured from the nose tip.
- Structural masses are computed by OpenRocket from the materials' densities. Items marked *est.* or *ASSUMED* are estimates: **weigh every part and update the file.**

Component	Material	Position (mm)	Length (mm)	Mass (g)
Nose cone - molded polystyrene plastic ogive for BT-80 (150 mm, 1.5 mm wall, 45 mm shoulder)	Polystyrene	0–150	150.0	51.3

Component	Material	Position (mm)	Length (mm)	Mass (g)
Upper body / payload tube - BT-80 kraft cardboard (66.0 OD x 65.0 ID)	Cardboard	150–550	400.0	29.9
Altimeter cradle - EPS foam puck with pocket (64.5 OD x 25 mm); 3 x 1.5 mm static vent holes in BT-80 here	Styrofoam (generic EPS)	195–220	25.0	1.4
Altimeter - Jolly Logic AltimeterTwo (ARC-approved), ~10 g	—	202–212	10.0	10.0
Egg capsule - 2-piece soft polyurethane foam (32 kg/m3), 2 egg-shaped cavities, 64.5 x 150 mm (11.5 g)	—	220–370	150.0	11.5
Raw egg #1 (59 g model; 55-63 g allowed)	—	235–292	57.0	59.0
Raw egg #2 (59 g model; 55-63 g allowed)	—	298–355	57.0	59.0
Ballast cartridge - 3D-printed PETG cup + twist-lock lid (63 OD x 30 mm), empty ~12 g (est.)	Cardboard	370–400	30.0	12.0
ADJUSTABLE BALLAST - modeling clay / printed PLA slugs in cartridge	—	373–397	24.0	61.0
Payload-floor bulkhead - kraft phenolic 3.2 mm, glued; Kevlar loop through 2 holes (no eyebolt)	Kraft phenolic	400–403	3.2	10.1
Parachute - ripstop nylon flat circle, 8 braided-nylon lines	Ripstop nylon	408–453	45.0	15.4
Shock cord (upper) - 1/4 in flat elastic 1.5 m, tied to bulkhead Kevlar loop & chute	Elastic cord (flat 6 mm, 1/4 in)	453–473	20.0	6.5
Nomex chute protector 9x9 in (~8 g)	—	473–483	10.0	8.0
Nose-cone tether - Kevlar thread 0.3 m + glue (1 g)	—	190–195	5.0	1.0
Glue + primer/paint allowance, nose & upper body (8 g, ASSUMED)	—	350–400	50.0	8.0
Launch lug - kraft paper, 1/4 in rod, 76 mm	Cardboard	464–540	76.0	1.3
Transition BT-80 -> BT-60 - molded polystyrene plastic, 1.5 mm wall	Polystyrene	550–615	65.0	31.4
Lower body / booster - BT-60 kraft cardboard (41.6 OD x 40.5 ID)	Cardboard	615–765	150.0	7.0
Motor mount tube - 24 mm kraft phenolic (24.1 ID x 25.7 OD)	Kraft phenolic	651–765	114.0	6.8
Centering ring aft - kraft phenolic 3.2 mm (40.5 OD / 25.7 ID)	Kraft phenolic	762–765	3.2	2.3
Centering ring fwd - kraft phenolic 3.2 mm (40.5 OD / 25.7 ID)	Kraft phenolic	651–654	3.2	2.3
Motor retainer - 3D-printed PETG screw-on (glued collar + cap), no metal (~8 g, est.)	—	753–771	18.0	8.0
Engine block / thrust ring - kraft paper, 24 mm (fwd end of motor)	Cardboard	692–698	6.0	0.7
Kevlar anchor: loop around MMT under fwd centering ring + epoxy (2 g)	—	651–659	8.0	2.0
Shock cord (lower) - Kevlar 1/8 in, 0.9 m, anchored to MMT	Kevlar 12-strand (3.2 mm, 1/8 in)	620–640	20.0	8.7
Glue fillets + primer/paint allowance, lower body & fins (8 g, ASSUMED)	—	705–755	50.0	8.0
Fins - 3 x balsa, through-the-wall to MMT, rounded edges	Balsa	635–765	130.0	18.5
Motor: AeroTech F44W-8 (single-use 24 × 70 mm)	—	698–768 (3 mm overhang)	70	48.0 (burnout 28.3)
Total at liftoff (baseline 61 g ballast)			**765**	**489.2**

Mass notes (assumptions):

- Glue/primer/paint allowance: 8 g upper + 8 g lower (*ASSUMED*).
- PETG cartridge: 12 g, PETG retainer: 8 g (*est.*).
- Foam egg capsule: computed for 32 kg/m³ foam.
- Plastic nose (OpenRocket: 51 g) and transition (31 g) are modeled as 1.5 mm polystyrene. **Real commercial parts may be lighter.**
- **If the as-built nose cone is lighter than modeled, press the difference in clay into the nose tip** to keep the CG where the model has it, then re-check the ballast with a test flight.

5.1 Fins

- **3 × trapezoidal ("clipped delta") balsa fins, 3/16 in (4.76 mm) thick.**
- Root chord 130 mm, tip chord 65 mm, span (height) 69.75 mm (use 70 mm), leading-edge sweep 65 mm. The trailing edge is straight (perpendicular to the body).
- Area 68.0 cm² each. Through-the-wall tab is 120 mm long × 7.4 mm high, glued to the motor tube. The fins sit flush with the aft end of the BT-60.
- The fin set weighs about 18.5 g plus fillets.
- Round the leading edge and tip, and lightly round the trailing edge. Glue fillets go at the body joint.
- **Flutter check** (NACA TN-4197 simplified method; balsa shear modulus G is uncertain, so three values are shown). Max velocity across all validation flights is 246–261 ft/s (Mach ≤ 0.245).
    - Flutter speed: about 480 ft/s (G = 0.05 GPa, pessimistic), 680 ft/s (0.10 GPa, nominal), 960 ft/s (0.20 GPa).
    - That is a **≥ 1.8× margin even in the pessimistic case.**
    - Optional: 1/8 in birch ply fins would be about 900 ft/s but heavier and aft. Rebalance the design if you switch.

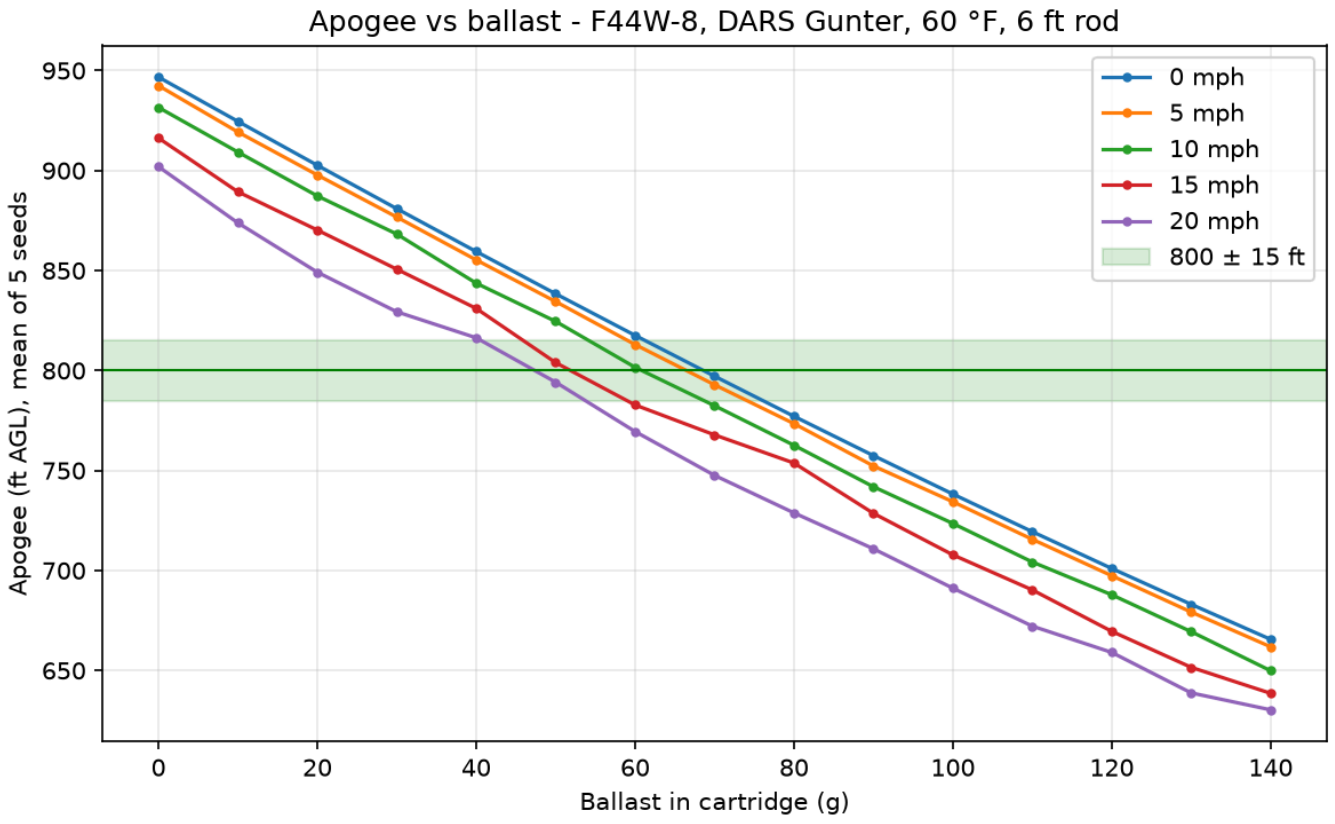
5.2 Recovery and egg protection

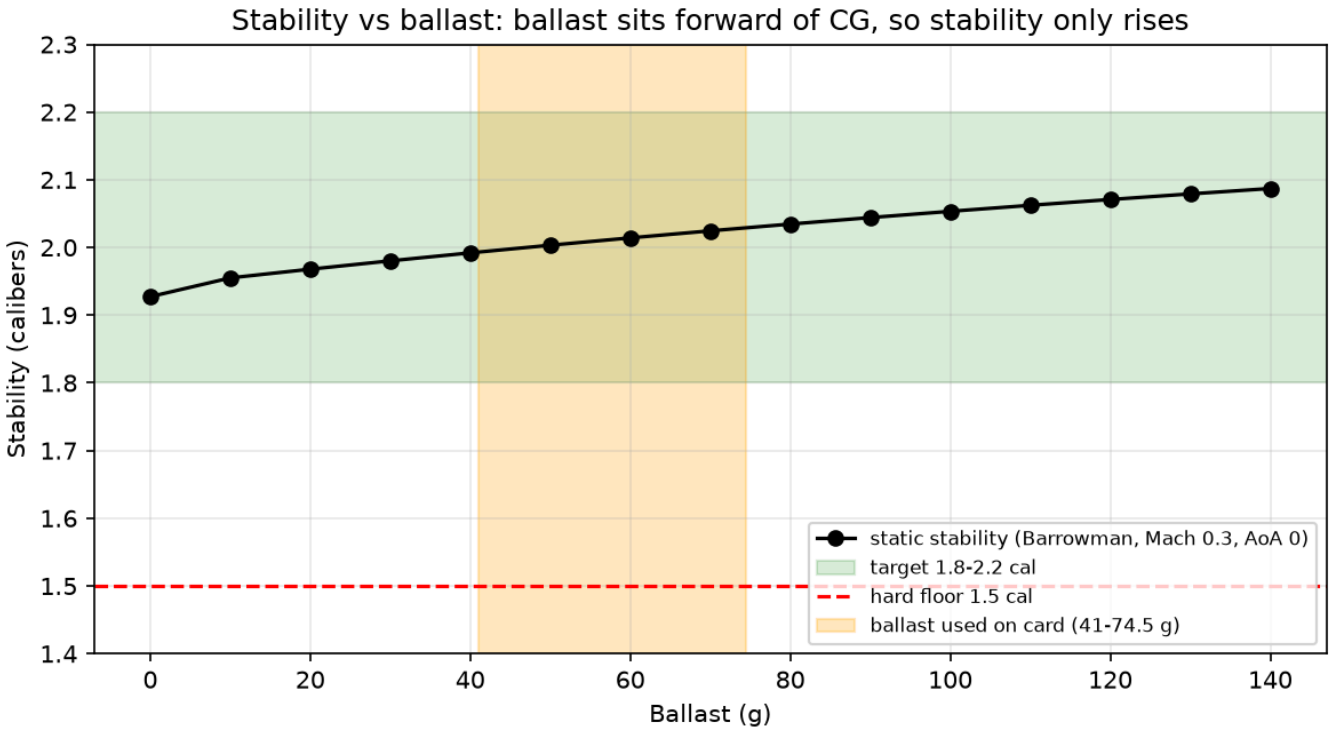
- **Parachute: 18.5 in (470 mm) flat ripstop nylon circle.**
    - Cut a 19.25 in circle and hem it 3/8 in, or buy a close size and tune it.
    - 8 lines of 2 mm braided nylon, each 18.5 in long, knotted to the elastic shock cord (no metal swivel). OpenRocket assumes CD 0.8.
    - Each 0.5 in of chute diameter changes flight time by about 0.7 s (Section 10).
- **Shock cords:** 1.5 m of 1/4 in flat elastic on the upper side and 0.9 m of 1/8 in Kevlar tied around the motor tube under the forward centering ring. Use a 9 × 9 in Nomex protector.
- **Egg capsule:** two-piece soft polyurethane foam, 64.5 mm diameter × 150 mm, with egg-shaped cavities.
    - Seat the eggs pointy-end-down with about 5 mm of foam all around. A thin plastic-wrap layer is fine but must be removed before inspection.
    - The capsule rests on the ballast cartridge lid, which rests on the glued bulkhead. The altimeter puck sits above it under the nose shoulder.
- **Loads on the eggs:**
    - Liftoff: peak acceleration 12.5–13.1 g.
    - Chute opening: 2.7–8.7 g. This is OpenRocket's instant-inflation estimate, so the real value is probably lower.
    - Landing: 23–26 ft/s vertical. With 20 mph wind, ground-relative speed is up to about 39 ft/s.
- **Altimeter:** drill 3 × 1.5 mm vent holes, 120° apart, in the BT-80 at the altimeter puck (about 200–212 mm from the nose tip).

6. Adjustable ballast system

- **Hardware:** a 3D-printed PETG cup, 63 mm OD × 30 mm tall, with 1.6 mm walls and a twist-lock (bayonet) lid. It slides into the BT-80 and sits on the 3.2 mm kraft phenolic payload bulkhead (glued at 400 mm). The egg capsule sits on its lid.
    - Inside volume is about 75 cm³. At an *assumed* 1.6–1.8 g/cm³ for oil-based modeling clay, that holds about 120–135 g, almost twice the largest card value (74.5 g).
    - Printed PLA/PETG slugs (about 1.25 g/cm³) also work up to about 90 g.
    - No metal.
- **Why it can't hurt stability:**

    - The ballast center is at 385 mm; the rocket CG is at 418–424 mm. The ballast is always forward of the CG but very close to it.
    - So each gram moves the CG forward slightly: **+0.0012 cal per gram**.
    - Static stability from the saved file, by ballast:
    - 0 g → **1.942**
    - 41 g → 1.993
    - 61 g (baseline) → 2.015
    - 74.5 g → 2.029
    - 100 g → 2.053
    - 140 g → 2.087
    - **Minimum = 1.94 cal with an empty cartridge. Every ballast setting gives 1.94–2.09 cal, and the card range gives 1.99–2.03 cal.**
    - At burnout (motor 28 g), stability rises to 2.21 cal.
- **How to use it at the field:**
    1. Read the wind (average, mph) and temperature (°F).
    2. Look up the ballast card value and add the egg correction (118 g − weighed egg total).
    3. Weigh clay on a 0.1 g scale. Use pre-made pellets (for example a 40 g base puck plus 10/5/2/1/0.5 g pellets, each wrapped in plastic film).
    4. Press the clay into the **center bottom** of the cup so it can't shift. Lock the lid and add one wrap of tape.
    5. Load: cartridge on the bulkhead → egg capsule → altimeter puck → nose. Tape the nose joint.
    6. Re-weigh the whole rocket with the motor. It should match the card mass within ±2 g (Section 8 table).
- **Rules of thumb:**
    - **1 g of ballast ≈ 2.0 ft of apogee** (from the ballast sweep and the ±1 g sensitivity test).
    - More wind means less ballast: about 0.4 g per mph at 0–5 mph, rising to about 1.5 g per mph at 15–20 mph.
    - 10 °F warmer ≈ 2 g more ballast.





7. Ballast cards

7.1 Primary card: 18.5 in chute, vertical 6 ft rod (tuned on 10–30 seeds, validated on 30 fresh seeds per cell)

Each cell shows ballast g / validated mean apogee ft / mean flight time s.

Temp	0 mph	5 mph	10 mph	15 mph	20 mph
40 °F	**64.5** / 800 / 39.51	**62** / 801 / 39.74	**57** / 801 / 39.64	**49.5** / 799 / 39.71	**41** / 800 / 40.10
50 °F	**66.5** / 800 / 39.18	**64** / 801 / 39.41	**59** / 800 / 39.28	**52.5** / 798 / 39.28	**43.5** / 797 / 39.54
60 °F	**68.5** / 800 / 38.85	**66.5** / 800 / 39.00	**61** / 800 / 38.95	**53.5** / 798 / 39.00	**46** / 798 / 39.26
70 °F	**70.5** / 800 / 38.52	**68.5** / 800 / 38.69	**63** / 800 / 38.63	**55.5** / 800 / 38.80	**46.5** / 801 / 39.11
80 °F	**72.5** / 800 / 38.19	**70.5** / 800 / 38.37	**65** / 801 / 38.31	**57** / 802 / 38.58	**48.5** / 804 / 38.94
90 °F	**74.5** / 800 / 37.86	**72.5** / 799 / 38.04	**67** / 800 / 38.00	**59.5** / 799 / 38.10	**52.5** / 800 / 38.39

7.2 8 ft rod option, 60 °F (ballast changes only 0–2 g)

	0 mph	5 mph	10 mph	15 mph	20 mph
Ballast (g)	68.5	66.5	62.5	55.5	47.5
Rail exit (ft/s)	79.6	79.9	80.6	81.8	83.4
Rail-exit stability, mean / worst (cal)	2.07 / 2.07	1.72 / 1.63	1.48 / 1.37	1.27 / 1.17	1.14 / 1.04
Apogee / flight time	800 / 38.85	801 / 38.99	800 / 38.94	800 / 39.03	801 / 39.36

7.3 Alternative card: 18 in chute (use on cold, windy days)

This card was also fully validated (30 fresh seeds per cell). Each cell shows ballast g / mean flight time s. Flight times are about 0.7–0.8 s shorter than with the 18.5 in chute.

Temp	0 mph	5 mph	10 mph	15 mph	20 mph
40 °F	65 / 38.71	63 / 38.87	57.5 / 38.76	50 / 38.78	42.5 / 39.10
50 °F	67 / 38.39	65.5 / 38.47	59.5 / 38.46	52.5 / 38.50	43 / 38.87
60 °F	69.5 / 38.01	67 / 38.22	62 / 38.03	54 / 38.37	46 / 38.42

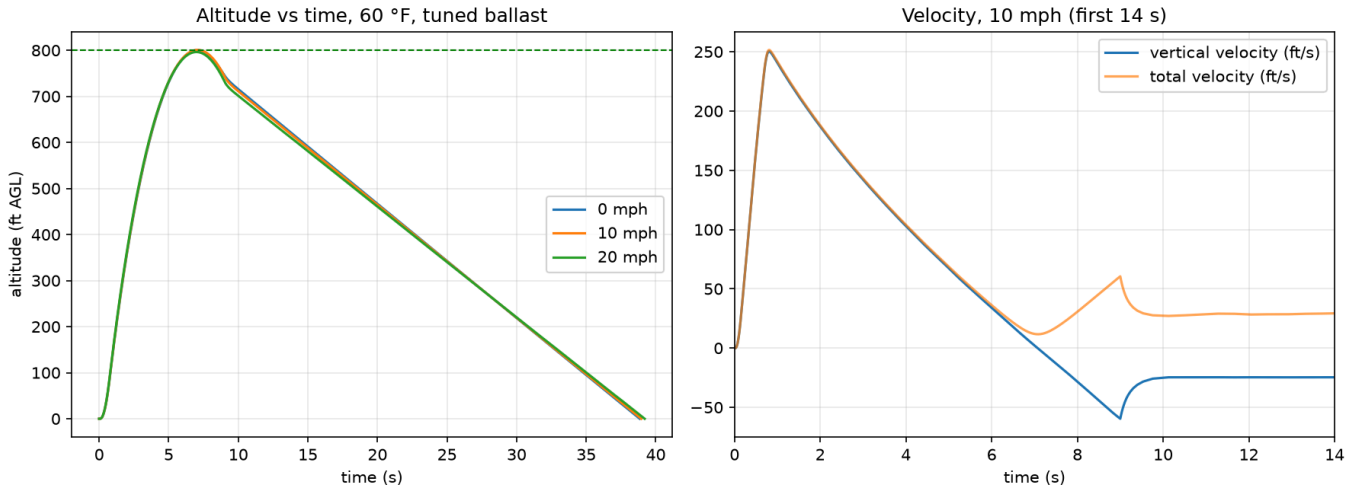
Temp	0 mph	5 mph	10 mph	15 mph	20 mph
70 °F	71.5 / 37.69	69.5 / 37.83	64.5 / 37.68	56.5 / 38.02	49 / 38.08
80 °F	73 / 37.42	71.5 / 37.52	66 / 37.41	58 / 37.70	50 / 37.98
90 °F	75 / 37.10	73 / 37.27	67.5 / 37.24	60.5 / 37.36	53 / 37.51

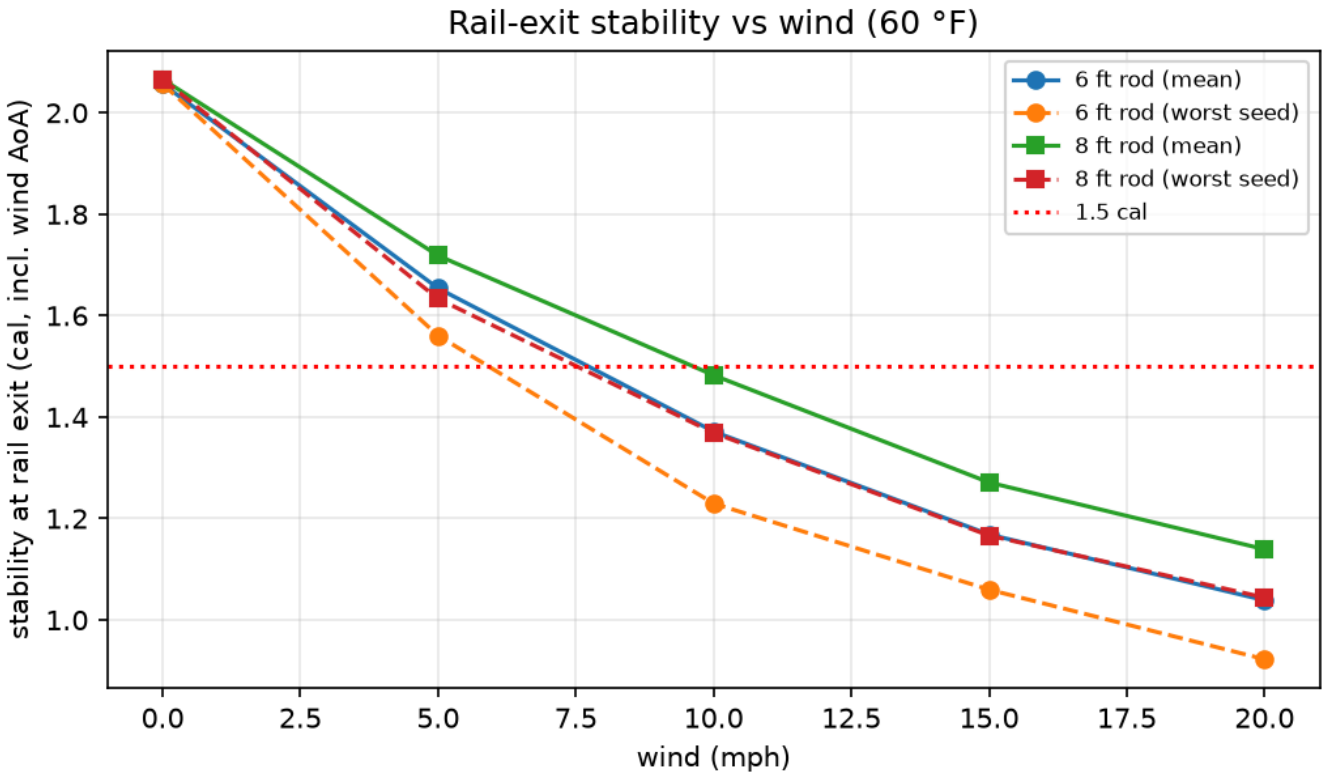
Chute rule:

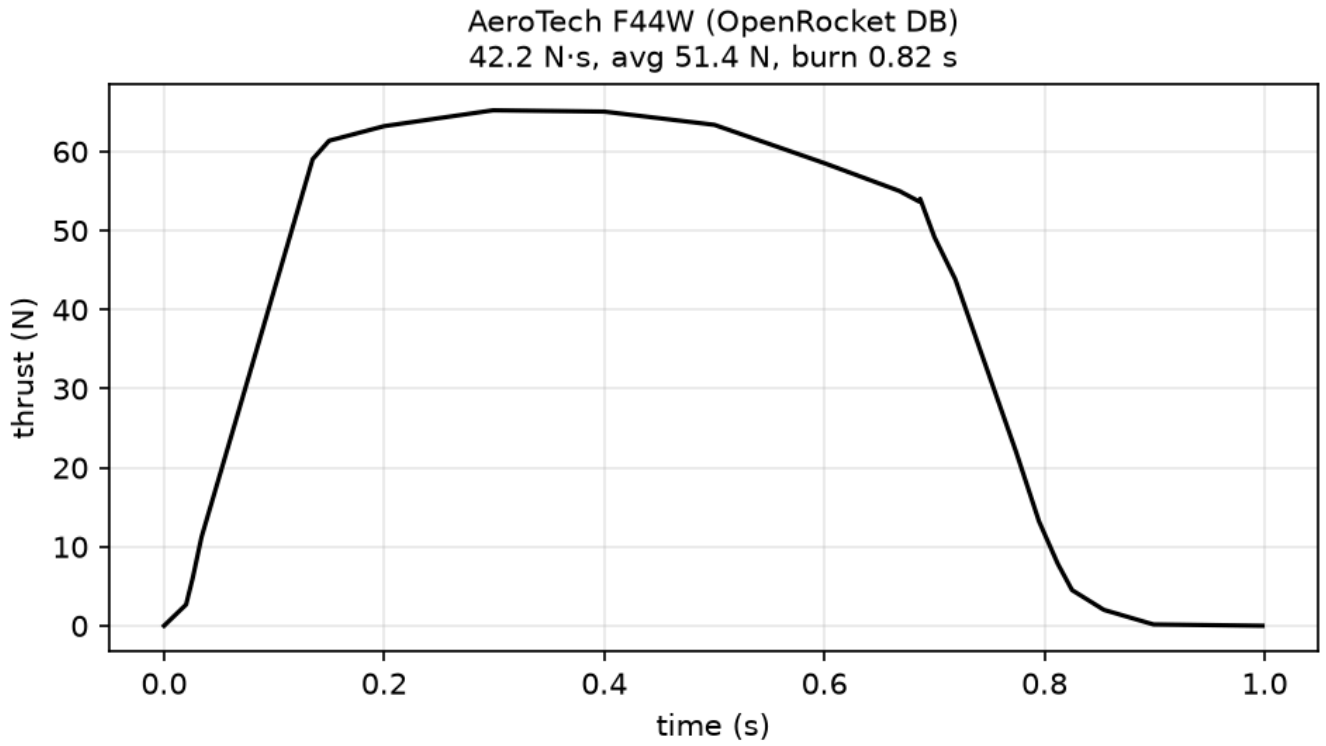
- Use **18.5 in** by default. It is centered on 38.2–39.5 s for 50–80 °F.
- Use **18 in** when it is **≤ 50 °F with ≥ 15 mph wind**, or **≤ 45 °F**. With 18.5 in, 40 °F and 15–20 mph averages 39.7–40.1 s.
- At ≥ 90 °F, the 18.5 in chute gives 37.9–38.4 s, while the 18 in chute would give only 37.1–37.9 s.

8. Key metrics (vertical 6 ft rod, DARS, card ballast)

Metric	Value
Length / max diameter	765 mm (30.1 in) / 66.04 mm (BT-80); ref. diameter 66.04 mm
Liftoff mass	489.2 g at baseline 61 g ballast; 469.2–502.7 g across the card
Empty mass (no motor, no ballast)	380.2 g
CG / CP (baseline)	CG 419.1 mm, CP 552.2 mm from nose tip (Barrowman, Mach 0.3)
Static stability	**1.94 cal (0 g) … 1.993–2.029 cal (card 41–74.5 g) … 2.09 cal (140 g)**; burnout 2.21 cal
Rail exit speed (6 ft / 8 ft rod)	66.6–71.3 ft/s / 79.6–83.4 ft/s
Rail-exit stability, 6 ft rod (mean, 60 °F)	0 mph 2.06 · 5 mph 1.65 · 10 mph 1.37 · 15 mph 1.17 · 20 mph 1.04 (worst seed 0.92)
Max velocity / Mach	246–261 ft/s / max Mach 0.245 (all 900 validation flights)
Peak boost acceleration	12.5–13.1 g
Time to apogee / ejection	about 7.0–7.1 s / about 9.0 s (burn 0.82 s + 8 s delay)
Deployment speed	48 ft/s (calm) – 65 ft/s mean (worst seed 68 ft/s, 20 mph)
Descent / landing speed (vertical)	23.4–25.5 ft/s (ground-relative incl. wind drift: up to 39 ft/s at 20 mph)
Drift (mean / worst seed)	0 mph about 9 ft · 5 mph about 153 · 10 mph about 295 · 15 mph about 463 · 20 mph about 650 ft (worst 769 ft)
Fin flutter margin	≥ 1.8× (pessimistic balsa G)







9. Per-condition results (ballast tuned per cell, then validated on 30 fresh random seeds)

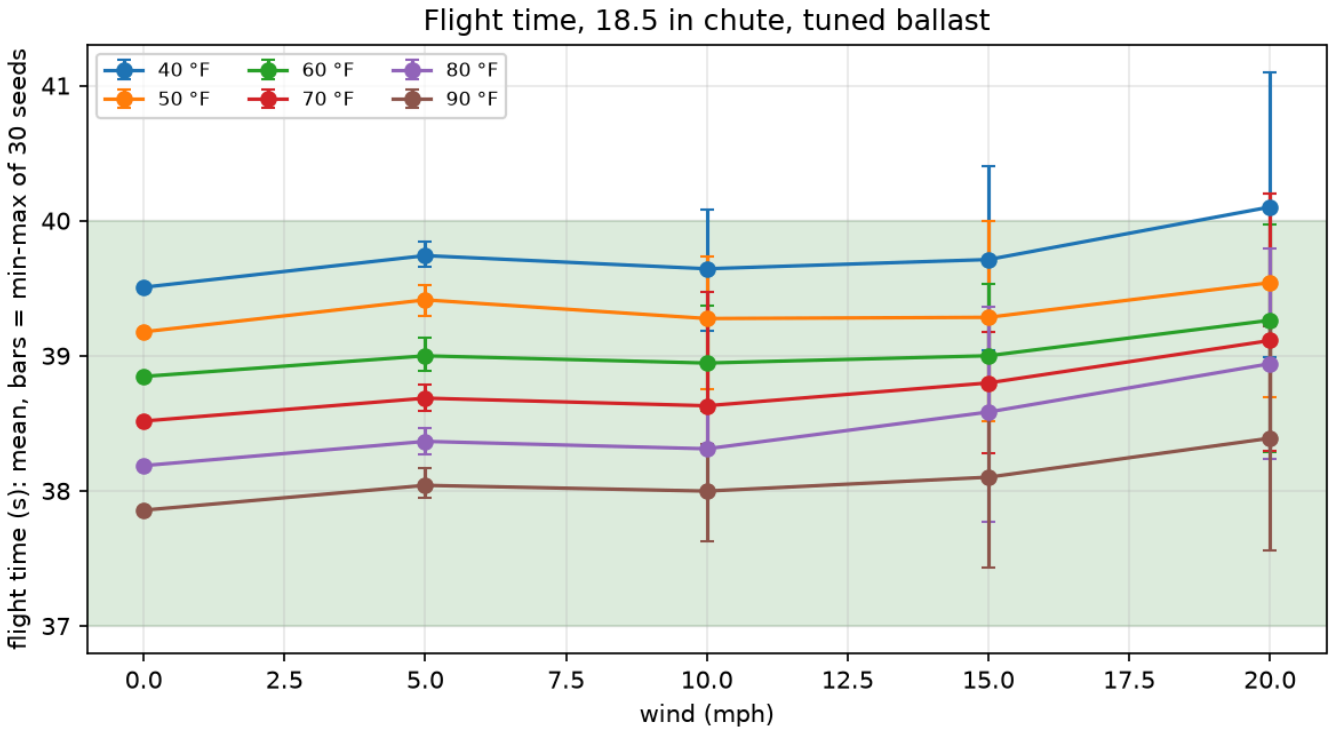
Column notes:

- "In" columns count how many of the 30 seeds landed inside 785–815 ft and inside 37–40 s.
- Rail-exit stability ("Rail stab") is shown as mean / worst seed and includes the wind angle at rail exit.

- Stability values are from the saved file's statics.

°F	mph	Ballast g	Mass g	Stab cal	Apogee mean (min–max)	In ±15	Flight time mean (min–max)	In 37–40	Rail exit ft/s	Rail stab	Deploy ft/s	Vert. land ft/s	Drift ft
40	0	64.5	492.7	2.019	799.6 (800–800)	30/30	39.51 (39.51–39.51)	30/30	68.0	2.05 / 2.05	48	24.1	9
40	5	62	490.2	2.016	800.6 (799–802)	30/30	39.74 (39.66–39.85)	30/30	68.3	1.66 / 1.54	50	24.0	164
40	10	57	485.2	2.011	800.7 (793–807)	30/30	39.64 (39.19–40.08)	28/30	68.9	1.42 / 1.29	60	23.8	315
40	15	49.5	477.7	2.003	798.8 (787–811)	30/30	39.71 (39.04–40.40)	23/30	70.0	1.19 / 1.07	63	23.6	479
40	20	41	469.2	1.993	800.5 (780–819)	28/30	40.10 (38.99–41.10)	13/30	71.3	1.07 / 0.95	64	23.4	684
50	0	66.5	494.7	2.021	800.0 (800–800)	30/30	39.18 (39.18–39.18)	30/30	67.7	2.05 / 2.05	48	24.4	9
50	5	64	492.2	2.018	801.1 (799–803)	30/30	39.41 (39.30–39.52)	30/30	68.0	1.66 / 1.58	50	24.3	159
50	10	59	487.2	2.013	800.4 (791–807)	30/30	39.28 (38.75–39.74)	30/30	68.7	1.38 / 1.23	61	24.2	309
50	15	52.5	480.7	2.006	797.6 (783–810)	29/30	39.28 (38.51–40.00)	29/30	69.6	1.20 / 1.03	63	23.9	477
50	20	43.5	471.7	1.996	796.7 (781–809)	27/30	39.54 (38.70–40.20)	23/30	70.9	1.04 / 0.92	65	23.6	660
60	0	68.5	496.7	2.023	800.2 (800–800)	30/30	38.85 (38.85–38.85)	30/30	67.4	2.06 / 2.06	48	24.7	9
60	5	66.5	494.7	2.021	800.0 (798–802)	30/30	39.00 (38.89–39.13)	30/30	67.7	1.65 / 1.56	50	24.6	153
60	10	61	489.2	2.015	800.4 (794–807)	30/30	38.95 (38.59–39.37)	30/30	68.4	1.37 / 1.23	60	24.4	295
60	15	53.5	481.7	2.007	798.0 (790–808)	30/30	39.00 (38.58–39.53)	30/30	69.4	1.17 / 1.06	63	24.2	463
60	20	46	474.2	1.999	798.2 (780–811)	28/30	39.26 (38.29–39.97)	30/30	70.5	1.04 / 0.92	64	23.9	650
70	0	70.5	498.7	2.025	800.1 (800–800)	30/30	38.52 (38.52–38.52)	30/30	67.1	2.06 / 2.06	48	25.0	8
70	5	68.5	496.7	2.023	800.3 (799–803)	30/30	38.69 (38.59–38.79)	30/30	67.4	1.66 / 1.58	50	24.9	156
70	10	63	491.2	2.017	800.3 (795–806)	30/30	38.63 (38.31–39.47)	30/30	68.1	1.37 / 1.27	60	24.7	291
70	15	55.5	483.7	2.009	800.3 (791–807)	30/30	38.80 (38.28–39.18)	30/30	69.2	1.17 / 1.08	63	24.5	464
70	20	46.5	474.7	1.999	800.6 (785–821)	29/30	39.11 (38.30–40.20)	29/30	70.5	1.04 / 0.94	64	24.2	645
80	0	72.5	500.7	2.027	799.9 (800–800)	30/30	38.19 (38.19–38.19)	30/30	66.8	2.06 / 2.06	48	25.2	8
80	5	70.5	498.7	2.025	800.0 (798–801)	30/30	38.37 (38.27–38.47)	30/30	67.1	1.66 / 1.55	50	25.2	149
80	10	65	493.2	2.019	800.6 (795–805)	30/30	38.31 (38.00–38.61)	30/30	67.9	1.38 / 1.27	60	25.0	285
80	15	57	485.2	2.011	802.1 (786–817)	29/30	38.58 (37.77–39.36)	30/30	69.0	1.20 / 1.01	62	24.7	458
80	20	48.5	476.7	2.002	803.6 (790–820)	28/30	38.94 (38.24–39.80)	30/30	70.2	1.09 / 0.94	63	24.5	650
90	0	74.5	502.7	2.029	799.6 (800–800)	30/30	37.86 (37.86–37.86)	30/30	66.6	2.06 / 2.06	48	25.5	8
90	5	72.5	500.7	2.027	799.4 (798–801)	30/30	38.04 (37.95–38.17)	30/30	66.8	1.65 / 1.57	50	25.5	143
90	10	67	495.2	2.021	800.4 (793–806)	30/30	38.00 (37.63–38.34)	30/30	67.6	1.38 / 1.25	60	25.3	284
90	15	59.5	487.7	2.014	798.9 (786–809)	30/30	38.10 (37.43–38.58)	30/30	68.6	1.18 / 1.06	63	25.0	437
90	20	52.5	480.7	2.006	800.0 (784–816)	28/30	38.39 (37.56–39.22)	30/30	69.6	1.08 / 0.93	63	24.8	634

Why some seeds miss: at 15–20 mph, OpenRocket's 15 % turbulence alone scatters apogee by about ±5–9 ft (1σ). At high wind a few single flights (up to 3 of 30) fall outside ±15 ft even with perfect ballast.



9.1 Saved simulations in the .ork file (60 °F, 6 ft rod, seed 1001 at save time)

Simulation name	Apogee (ft)	Flight time (s)	Rail exit (ft/s)	Rail stab	Deploy (ft/s)	Landing (ft/s)	Drift (ft)
DARS 0 mph, 60F, 6 ft rod - ballast 68.5 g (+7.5 g vs baseline)	800.2	38.85	67.4	2.06	48.1	24.7	9
DARS 5 mph, 60F, 6 ft rod - ballast 66.5 g (+5.5 g vs baseline)	799.3	38.95	67.7	1.61	51.1	25.7	149
DARS 10 mph, 60F, 6 ft rod - ballast 61 g (+0 g vs baseline)	800.7	38.96	68.4	1.43	60.5	27.8	324
DARS 15 mph, 60F, 6 ft rod - ballast 53.5 g (-7.5 g vs baseline)	795.2	38.88	69.4	1.07	64.1	33.2	458
DARS 20 mph, 60F, 6 ft rod - ballast 46 g (-15 g vs baseline)	796.3	39.20	70.5	1.01	64.9	41.3	591

Notes on the saved file:

- OpenRocket uses whatever ballast mass is currently in the file (the baseline, 61 g) when you re-run a simulation. **Set the "ADJUSTABLE BALLAST" mass to the grams in the simulation name before re-running it.**
- OpenRocket does not store the random seed in the .ork. The calm (0 mph) simulation re-runs identically; windy re-runs give a new turbulence pattern, which scatters apogee by about ±3–9 ft (1σ).
- The file was re-opened and checked: motor, configuration and ballast are correct, the calm re-run matched to 0.01 ft, and the stored results are intact.

10. Sensitivity and Monte Carlo

10.1 One-at-a-time sensitivity (60 °F, card ballast, 10 seeds each; at 10 mph, turbulence-sampling noise is about ±5 ft, so changes under about 5 ft there are not meaningful)

Change	0 mph Δapogee (ft)	0 mph Δtime (s)	10 mph Δapogee (ft)	10 mph Δtime (s)
impulse x0.95	-60.4	-2.89	-57.0	-2.88
impulse x1.05	+60.4	+2.83	+62.6	+3.11
rocket CD x0.9	+22.4	+1.14	+26.5	+1.45
rocket CD x1.1	-20.9	-1.07	-19.0	-1.06
build mass -10 g	+16.4	+1.04	+19.7	+1.25
build mass +10 g	-20.1	-1.26	-16.5	-1.07
ballast -1 g	+2.0	+0.13	+5.5	+0.32
ballast +1 g	-2.0	-0.13	+1.4	+0.05
eggs 55 g each (uncompensated)	+16.4	+1.02	+19.3	+1.24
eggs 55 g each, ballast +8 g (compensated)	-0.0	-0.01	+4.9	+0.29
eggs 63 g each (uncompensated)	-16.1	-0.99	-12.2	-0.83
eggs 63 g each, ballast -8 g (compensated)	+0.0	+0.01	+2.4	+0.14
chute CD 0.72	+0.0	-1.53	+4.7	-1.25
chute CD 0.88	+0.0	+1.46	+4.3	+1.71
temperature 50F	-4.2	+0.08	-3.0	+0.11
temperature 70F	+4.1	-0.08	+6.9	+0.09
8 ft rod	-0.0	+0.00	+4.5	+0.26
rod 5 deg dir 0	-6.2	+0.05	-2.9	-0.21
rod 5 deg dir 90	-6.2	+0.04	-17.8	-1.00
rod 5 deg dir 180	-6.2	+0.05	-3.2	-0.23
rod 5 deg dir 270	-6.2	+0.06	+11.8	+0.77

- "rod 5 deg dir …" is the rod tilt direction (compass degrees). In these runs OpenRocket's wind comes from 90°.
- At 10 mph, tilting 5° **toward the wind** (dir 90) cost about 18 ft, because the rocket weathercocks even further. Tilting 5° **away from the wind** (dir 270) added about 12 ft. In calm air any 5° tilt cost about 6 ft.
- **Keep the rod vertical (the baseline), or record the tilt and correct for it.** The RSO decides any rod tilt.

Ejection-delay tolerance (real delays vary from nominal). Each cell shows mean flight time s / mean deploy speed ft/s.

Actual delay	0 mph	10 mph	20 mph
7 s	39.65 / 30	40.06 / 31	40.76 / 35
8 s	38.85 / 48	39.03 / 59	39.16 / 65
9 s	37.73 / 58	36.57 / 90	36.82 / 92
10 s	36.78 / 59	33.37 / 116	33.46 / 118

A delay running 1 s **long** pushes deployment to about 58–94 ft/s and flight time to 36.6–37.7 s. **Record the ejection time on video during the test flight.**

Chute size vs. flight time (mean of 10 seeds; card ballast from the 18 in tuning). Each row lists 50 / 60 / 70 / 80 °F × (0, 10, 20 mph):

- **17 in:** 36.92, 36.94, 37.33, 36.56, 36.61, 36.97, 36.25, 36.27, 36.58, 36.01, 36.15, 36.57 (range 36.01–37.33 s)
- **18 in:** 38.39, 38.56, 38.99, 38.01, 38.07, 38.64, 37.69, 37.72, 38.08, 37.42, 37.47, 37.76 (range 37.42–38.99 s)

- **18.5 in:** 39.11, 39.25, 39.79, 38.72, 38.71, 39.35, 38.39, 38.43, 38.96, 38.12, 38.18, 38.56 (range 38.12–39.79 s)
- **19 in:** 39.83, 39.83, 40.83, 39.43, 39.50, 39.81, 39.09, 39.14, 39.53, 38.82, 38.85, 39.49 (range 38.82–40.83 s)
- **20 in:** 41.25, 41.36, 41.85, 40.83, 40.91, 41.54, 40.47, 40.56, 40.70, 40.18, 40.22, 40.63 (range 40.18–41.85 s)

10.2 Monte Carlo (60 °F, card ballast, 200 runs per wind, each run with a new turbulence seed)

A: before calibration. All inputs drawn uniformly over the requested ranges:

- motor impulse ±5 %
- build mass ±10 g
- ballast ±1 g
- rocket CD ±10 %
- chute CD ±10 %
- launch angle 0–5° in a random direction

B: after test-flight calibration. Average CD and mass biases are removed (CD ±2 %, mass ±2 g, chute CD ±3 %). Motor-to-motor impulse ±5 % stays, because it changes from motor to motor.

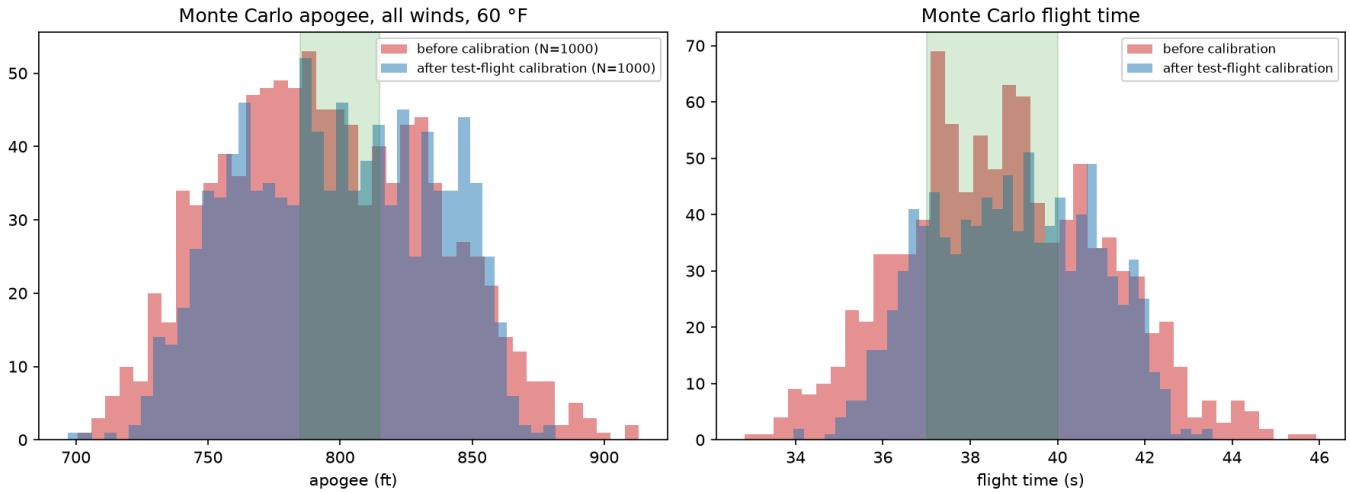
C: same as B but with no impulse variation. This shows what ballast tuning alone can control.

Case	Wind	N	Apogee mean ± SD (ft)	P5–P95 (ft)	In 785–815	Flight time mean ± SD (s)	P5–P95 (s)	In 37–40
A pre-cal	0	200	795 ± 39	733–859	28%	38.66 ± 2.15	35.03–42.13	48%
A pre-cal	5	200	794 ± 37	738–855	26%	38.66 ± 2.14	35.30–42.22	52%
A pre-cal	10	200	797 ± 39	737–862	22%	38.75 ± 2.23	35.19–42.22	46%
A pre-cal	15	200	797 ± 44	732–867	22%	38.81 ± 2.50	35.13–43.51	44%
A pre-cal	20	200	795 ± 40	728–857	26%	39.06 ± 2.31	35.12–42.74	48%
B post-cal	0	200	802 ± 32	746–851	31%	38.96 ± 1.57	36.34–41.31	58%
B post-cal	5	200	801 ± 36	745–855	28%	38.97 ± 1.78	36.31–41.81	50%
B post-cal	10	200	798 ± 36	743–854	28%	38.86 ± 1.87	35.82–41.86	49%
B post-cal	15	200	800 ± 36	744–855	22%	39.12 ± 1.86	36.30–41.91	48%
B post-cal	20	200	792 ± 40	729–855	25%	38.91 ± 2.08	35.51–42.09	48%
C post-cal, no impulse var.	0	100	798 ± 4	791–806	100%	38.84 ± 0.38	38.10–39.46	100%
C post-cal, no impulse var.	10	100	800 ± 7	787–810	96%	38.94 ± 0.49	38.21–39.69	98%
C post-cal, no impulse var.	20	100	796 ± 14	774–814	77%	39.16 ± 0.80	37.76–40.27	82%

What this means:

- **Motor impulse variation is the dominant uncertainty.** ±5 % impulse is about ±60 ft of apogee and about ±2.9 s of flight time. Drag (±10 % CD → about ±22 ft) and build mass (±10 g → about ±18 ft) are next.
- **The test flight removes the drag and mass biases.** Motor-to-motor scatter remains.

- With impulse variation removed (case C), **96–100 % of flights at 0–10 mph and 77 % at 20 mph land within 800 ± 15 ft**, and 82–100 % land in the 37–40 s window.
- **To reduce motor scatter:** buy all motors from one production lot (same date code), fly the calibration flight on that lot, and keep motors at similar temperatures before flight.
- **The ±5 % impulse range is the requested test bound, not measured lot data.**



11. Risks and thin margins (honest list)

1. **Rail-exit stability in wind** (OpenRocket value, which includes the wind angle at rail exit): 1.37 cal at 10 mph, 1.17 at 15 mph, 1.04 at 20 mph (worst seed 0.92).
    - This is **below 1.5 cal at ≥ 10 mph** even though static stability is 2.0 cal.
    - It recovers fast as speed builds: in the 10 and 20 mph sample flights, stability passed 1.5 cal within 0.03–0.09 s after leaving the rod and was about 1.7–2.0 cal by 0.15 s.
    - Mitigations: use an 8 ft rod (1.48 / 1.27 / 1.14 cal), or fly in the calmer morning.
    - Static margin was **not** raised further, because that would increase weathercocking and the apogee loss in wind.
2. **Late deployment (8 s delay):** about 1.9 s after apogee at 48–65 ft/s, with an OpenRocket opening load of up to about 8.7 g in 20 mph wind.
    - A long-running delay is the bigger risk (see the delay table).
    - Mitigations: elastic cord + Kevlar, a well-packed chute, and the foam egg capsule. Inspect the shock cords after every flight.
3. **Motor impulse scatter** of about ±5 % ≈ ±60 ft is the largest apogee risk. See the Monte Carlo above; use one motor lot.
4. **Flight time near the window edges:**
    - Cold + windy (≤ 50 °F, ≥ 15 mph): about 39.3–40.1 s with 18.5 in; use the 18 in card.
    - Hot (≥ 90 °F): 37.9 s.
    - Real chute CD is unknown: ±10 % ≈ ±1.5 s. **The test flight must measure it.**
5. **As-built masses:** the plastic nose and transition masses are model estimates, and glue/paint is assumed. A lighter nose moves the CG aft (about −0.01 cal per gram removed from the nose tip region). **Weigh the parts; add nose-tip clay if needed to match the model CG of 419 mm ± 3 mm at baseline.**
6. **BT-80 OD is exactly 2.600 in nominal, which is right at the rule minimum.** Measure the tube you buy.
7. **High-wind drift:** about 650 ft mean (up to about 770 ft) at 20 mph. Check the DARS field size and downrange hazards.
8. **Finals use rails.** Adding printed 1010 rail guides adds drag. Re-simulate and re-tune the ballast before Finals; the target there is 775–825 ft.

9. **Model limits:**
    - OpenRocket does not model motor temperature effects. Real motors are slightly weaker when cold; calibrate in the expected temperature.
    - Barrowman/OpenRocket stability at large angles of attack is approximate.
    - Parachute opening is modeled as instantaneous.

12. Test-flight plan: DARS Gunter, Sat Oct 17, 2026 (test, not scored)

Climate context: NOAA 1991–2020 daily normals for Oct 17 are a high of 77.6 °F, low of 53.8 °F (McKinney Muni AP) and a high of 75.0 °F, low of 54.7 °F (Sherman). Expect roughly 60–78 °F during launch hours.

Before the launch:

1. Build the rocket and weigh every part. Update the .ork masses (nose, transition, tubes, fins, cartridge, retainer). Check that the CG with motor and baseline ballast is within ±3 mm of 419 mm. Do a swing test or balance test.
2. Print the cartridge and retainer, and test-fit the twist-lock lid. Make pre-weighed clay pellets.
3. Pack the chute and practice the load sequence. Bring the 18 in and 18.5 in chutes and the scale.
4. Buy fresh F44W-8 motors (post-Aug-2024) from a single lot, and record the lot/date code.

At the field:

1. Record the wind (handheld anemometer: average and gust, mph), temperature, and pressure/humidity if available.
2. Weigh the eggs and set the ballast from the card plus the egg correction. Weigh the whole rocket with the motor and record it.
3. Use a vertical 6 ft (or 8 ft) rod. Note the rod length and any tilt.
4. Video the flight from the side, and use two stopwatches for flight time. Record:
    - liftoff time
    - ejection time (from video)
    - landing time
    - altimeter apogee (AltimeterTwo also gives max speed and acceleration)
    - whether the eggs survived
    - drift distance and direction
5. Inspect after landing: egg condition, shock cord and chute burns, fin and joint damage, motor retainer.

Recalibration after the flight:

1. **Model check.** In OpenRocket, set the measured conditions (wind, temperature, rod, *as-flown mass/ballast*). Run about 10 sims and take the mean predicted apogee **A_pred**.
2. **Ballast correction.** Δ = A_meas − A_pred. Add **Δ / 2.0 g** of ballast to every card cell. For example, flying 830 ft when 800 ft was predicted → add 15 g. (Apogee slope is about 2.0 ft per gram near 800 ft.)
3. **Optional: model correction.** To make the model match, change the rocket CD by about **−Δ/2.2 percent** (10 % CD ≈ 22 ft; flying high means the real drag is lower). Then re-tune the card in OpenRocket.
4. **Flight-time correction:**
    - Use the measured descent time = (landing time − ejection time).
    - New chute diameter ≈ D × (38.5 − t_eject) / (T_meas − t_eject), with t_eject ≈ 9.0 s. For example, if T_meas = 37.2 s with 18.5 in, the new D ≈ 18.5 × 29.5/28.2 ≈ 19.4 in.
    - Or remember: each 0.5 in of diameter ≈ 0.7 s.
5. **Delay check.** If the video shows ejection much later than about 9 s, expect higher deployment speeds. Consider reinforcing the shock cords.
6. Re-fly with the corrected card. Two consistent flights are better than one.

13. Simulation log

All runs used OpenRocket 24.12 with DARS Gunter conditions:

- location 33.438004 N, −96.803632 W, 219.86 m elevation
- standard pressure at the field (98712 Pa)
- temperatures 40/50/60/70/80/90 °F
- 15 % turbulence, vertical rod (6 ft baseline, 8 ft option)
- RK4 6-DOF with OpenRocket defaults

Stage	Script	Simulations
Speed / listener tests	t_speed, t_lst	11
Stage 1 motor screen (all AeroTech/Quest E–F)	screen.py	1,122
Airframe geometry study (14 variants, 6 & 8 ft rod)	geo_test.py	168
Stage 2 finalists (5 motors × 20 conditions, tune + validate)	stage2.py	3,580
Final ballast tuning + 30-seed validation: run 1 (18 in, 10-seed tuning), run 2 (18 in, 30-seed tuning for ≥ 10 mph), run 3 (18.5 in, final) — 6 temperatures × 5 winds + 8 ft rod	final_tune.py	16,240
Chute-size sensitivity	chute_sens.py	600
Ballast sweep 0–140 g	sweep.py	375
One-at-a-time sensitivity	oat.py	440
Ejection-delay tolerance	delay_sens.py	120
Monte Carlo A / B / C	mc, mc_post, mc_noimp	2,300
Saved sims + file verification	final_save, verify	10
Total		**24,970**

14. Assumptions (all labeled; replace with measured values)

- **Masses:**
  - Glue/primer/paint 8 g + 8 g; PETG cartridge 12 g; PETG retainer 8 g; Nomex 8 g; Kevlar anchor 2 g; nose tether 1 g.
  - Altimeter 10 g; eggs 59 g each.
  - Foam capsule density 32 kg/m³.
  - Plastic nose and transition modeled as 1.5 mm polystyrene.
- **Chute:** CD 0.8 for a flat ripstop canopy.
- **Clay density:** 1.6–1.8 g/cm³ (for cartridge capacity only).
- **Balsa shear modulus:** 0.05–0.20 GPa (flutter only).
- **Motor:** OpenRocket database curve for F44W (42.2 N·s vs. 41.5 N·s on the ARC list); no temperature dependence.
- **Weather:** wind modeled as constant average speed with 15 % turbulence; standard pressure at field elevation.
- **Source files:** the design scripts and all raw results (JSON/CSV) are in `/workspace/rocket/aerotech_new/` on the analysis computer.